

FinReflectKG — Time-Travel demo visualizer

A lightweight, **local**, **live** demo of the FinReflectKG time-travel layer (G9/§4.8), served against the `FinReflectKgTemporal` database. Fresh build (FastAPI + Cytoscape.js), no framework build step.

What it shows (v1.5)

- **DVR time-slider** — scrub 2014→2024 and the company's subgraph **crossfades** between years, DVR-style. The whole decade arrives in one `/api/timeline` payload; the union of all years is laid out **once** with stable positions, so scrubbing is pure client-side (no fetch, no relayout). The company node is **pinned at the centre**; facts fade in / out as they appear / disappear.
- **Depth** control (1 = direct-facts star / 2–3 = connected context). Depth ≥ 2 adds genuine concept↔concept edges, and the header reports how many — `depth 2 (20 context)` — because a company node is incident to ~99 % of its own neighbourhood, so without a reserved budget the star swallows every slot and the deeper views are merely denser stars. Each context edge is taken with the shortest path anchoring it to the company, so the view is always a *connected* neighbourhood — no free-floating islands.
- **Valid / Reported time toggle** — *valid* = what **held** at mid-year (`validFrom / validTo`); *reported* = what the filing **asserted** that year (`year` , transaction time). The bitemporal (P3) axis, made interactive.
- **Legend = type filters** — click a legend swatch to show / hide that entity type.
- **Top-N PageRank filter** (Influence panel) — keeps only entities ranked in the **corpus-wide** top-N at the current anchor year, *not* the top-N within this company. The global leaders are hubs like `net income` and `united state` , so only a fraction land in any one company's neighbourhood: the control reads `global PageRank top 200 – 20 of 219 here` so the survivor count is explicit rather than looking like a broken filter.
- **Influence over time** — top entities by GAE PageRank at the anchor year nearest the slider (`gae_pr_2014/2019/2020/2024`).
- **Company explorer + diffs** — year-over-year *appeared / disappeared* facts (vs 2014) and **backward-looking disclosures** (facts a filing asserts about periods ≥ 3 years earlier).

Run

```
# from the repo root (connection comes from .env; DB = FinReflectKgTemporal)
.venv/bin/python -m uvicorn demo.api:app --port 8080
# then open http://localhost:8080 (try tickers: aapl, msft, amzn, ... any of the 743)
```

Endpoints (backend)

GET /api/years · GET /api/tickers · GET /api/timeline?ticker=&depth=&clean=&axis= (all years, one payload) · GET /api/asof?ticker=&year=&limit=&depth=&clean= · GET /api/influence?year=&top= · GET /api/pranks?year=&top= · GET /api/diff?ticker=&from=&to= · GET /api/backward?ticker=&lag=

Notes

- Read-only; uses the stdlib REST helper (`scripts/arango.py`) driven by `.env` .
- Cytoscape.js is **vendored** at `demo/static/vendor/cytoscape.min.js` — fully offline, no CDN.
- The as-of canvas shows a connected neighbourhood of the company (depth 1 = a clean star), capped at ~140 facts for readability and always island-free. The header shows *shown / total · depth N (M context) · as of YEAR*.
- **Theme** — light by default, with a dark toggle in the header (persisted per device). The OS `prefers-color-scheme` is deliberately not consulted, so a demo on a dark-mode laptop still opens light. Palette is the ArangoDB one shared with the r2g Studio; the Cytoscape canvas reads the CSS tokens and is restyled on toggle.
- Selection is **reproducible** — both selection queries carry a deterministic `$SORT e._key` . Without it the row order, and therefore the rendered picture, varied between runs.
- **Cleaned/Raw toggle** (header): Cleaned drops junk placeholders + skolemizes generic hubs to per-company bnodes (dashed green); Raw shows the graph as extracted (shared hubs + junk diamonds).